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Targeted RNA editing: novel tools to study post-transcriptional regulation

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SUMMARY

RNA binding proteins (RBPs) regulate nearly all post-transcriptional processes within cells. To fully understand RBP function, it is essential to identify their *in vivo* targets. Standard techniques for profiling RBP targets, such as crosslinking immunoprecipitation (CLIP) and its variants, are limited or suboptimal in some situations, e.g. when compatible antibodies are not available and when dealing with small cell populations such as neuronal subtypes and primary stem cells. This review summarizes and compares several genetic approaches recently designed to identify RBP targets in such circumstances. TRIBE (targets of RNA binding proteins identified by editing), RNA tagging, and STAMP (surveying targets by APOBEC-mediated profiling) are new genetic tools useful for the study of post-transcriptional regulation and RBP identification. We describe the underlying RNA base editing technology, recent applications, and therapeutic implications.

INTRODUCTION TO RNA EDITING

Adenosine-to-inosine (A-to-I) editing

RNA processing includes RNA modification. Modified RNA moieties include inosine, which is almost always the result of A-to-I editing. This is caused by deamination at ^{6}A and is the most prevalent type of RNA base editing in metazoan cells (Nishikura, 2010). A-to-I editing events on tRNAs are carried out by ADATs (adenosine deaminase that act on tRNA; Torres et al., 2014), whereas ADARs (adenosine deaminases that act on RNA) are solely responsible for A-to-I editing on mRNAs (Melcher et al., 1996) and other non-coding RNAs (Nishikura, 2016).

ADAR genes are highly conserved in metazoans. There are three different human ADAR genes: hADAR1, hADAR2, and hADAR3. The single *Drosophila* ADAR gene, dADAR, is most related to hADAR2 (Palladino et al., 2000). ADAR enzymes contain double-stranded RNA-binding motifs (dsRBMs) and a catalytic deaminase domain (referred to hereafter as ADARcd, with cd indicating catalytic domain). ADAR domains are modular and can work independently, with the dsRBMs being predominantly responsible for target recognition and the ADARcds for deamination (Figure 1; Liu et al., 2000; Nishikura, 2010).

Endogenous ADAR editing targets are structured and include regions with a large fraction of double-stranded character (Dawson et al., 2004). Upon target binding, ADARcd deaminates adenosines to inosines, which are recognized as guanosines by ribosomes as well as by reverse transcriptases (Figure 1). A-to-I editing events can therefore be identified by A-to-G mutations with standard RNA sequencing (RNA-seq) methods. The best-understood example of endogenous ADAR editing function is on GluA2 mRNA: its 607th codon CAG (Gln) is nearly 100% edited to CIG (Arg) in the mammalian CNS (Sommer et al., 1991). This causes an essential decrease in Ca^{2+} permeability of AMPA receptors assembled with GluA2 (Hollmann et al., 1991; Hume et al., 1991).

Cytosine-to-uracil (C-to-U) editing

Cytosine-to-Uracil editing (deamination at ^{4}C) is a conserved cross-kingdom RNA editing event, catalyzed by members of the APOBEC (apolipoprotein B mRNA editing enzyme, catalytic polypeptide-like) family (Salter et al., 2016). C-to-U editing was first discovered in apolipoprotein B (*apoB*) mRNA, in which a genomically absent C-to-U base change at codon 2153 ($Q^{CAA} \rightarrow X^{UAA}$) led to the production of the short apoB-48 isoform (Chen et al., 1987; Powell et al., 1987). The responsible enzyme, APOBEC1, was identified in 1993 (Teng et al., 1993) and subsequently joined by other members of the APOBEC family (Conticello, 2008; Davidson et al., 1995; Salter et al., 2016).

All members of the APOBEC family contain one or more zinc-dependent cytidine deaminase domain(s) (ZDD; Salter et al., 2016), which is the source of catalytic activity. APOBEC family members typically use both RNA and single-stranded DNA (ssDNA) as substrates, but several members only possess ssDNA editing ability (Salter et al., 2016). However, APOBEC1 has only low RNA affinity and is therefore unable to edit its substrates (e.g., *apoB* mRNA) without the presence of an RNA binding partner, for example APOBEC1 complementation factor (ACF; Blanc and Davidson, 2010; Smith et al., 2012). A minimal editing machinery is reported to require APOBEC1 dimers (Lau

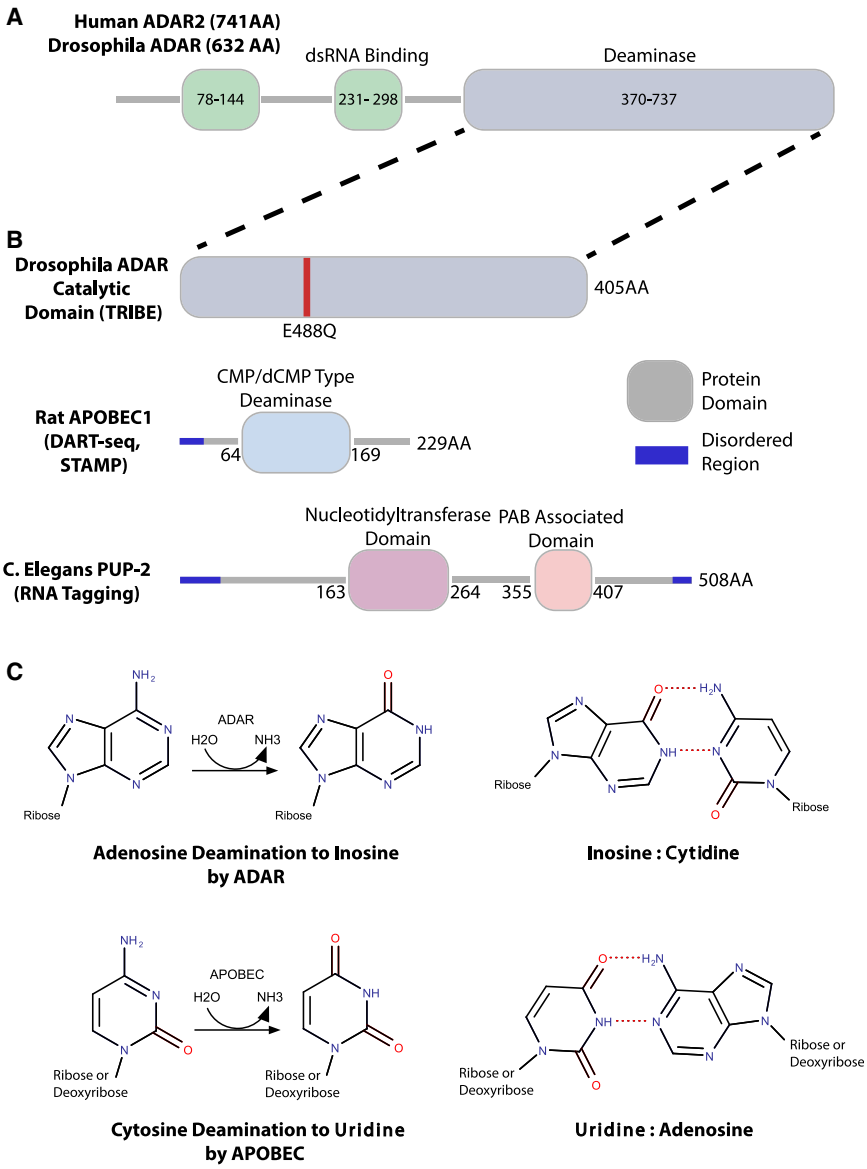

Figure 1. ADAR and other enzymes used to profile targets of RNA binding proteins (RBPs)

(A) Different human ADAR isoforms used for TRIBE. Green box, dsRNA binding domain; blue box, deaminase domain. Numbers correspond to amino acids of human ADAR2.

(B) Enzymes that are used to profile targets of RBPs. Boxes, protein domains; blue bars, disordered regions, red vertical ba, hyperactivating mutation of ADAR catalytic domain. Different domains and their amino acid positions are highlighted (numbers).

(C) Enzymatic conversion of nucleosides and effects on nucleoside pairing. Left, chemical structures of ADAR and APOBEC substrates. Arrows represent enzymatic conversion by ADAR (top) and APOBEC (bottom). Right, after conversion, nucleosides can hydrogen bond with different partners, leading to readout during cDNA synthesis, or changes in tRNA anti-codon binding.

et al., 1994; Navaratnam et al., 1998) and RNA-bridged ACF dimers, forming a 27S editosome (Sowden et al., 2002).

RNA EDITING CAN IDENTIFY RNA BINDING PROTEIN (RBP)-TARGET TRANSCRIPTS

RBPs

As the central molecule in gene expression, RNA carries out a diversity of functions, from the templating of protein synthesis by messenger RNAs (mRNAs; Crick et al., 1961; Nirenberg, 2004) to the hydrolytic activities of ribozymes (Cech, 2012; Guerrier-Takada et al., 1983; Kruger et al., 1982). All transcripts experience a variety of regulatory steps that occur during or after their transcription, namely co-transcriptional processing events or strictly post-transcriptional processing events that follow transcript release from the transcriptional machinery. RNA processing is now a robust field and has been investigated in recent years with increasingly sophisticated genetic, biochemical, and structural methods (Andersson and Sandelin, 2020; Biswas et al., 2021a; Lee and Rio, 2015; Shi, 2017; Zhang et al., 2018).

RBPs are fundamental to all aspects of RNA processing, including alternative splicing, polyadenylation, nuclear export, intracellular localization, degradation, and translation (Gerstberger et al., 2014; Jansen, 2001; Szostak and Gebauer, 2013; Witten and Ule, 2011; Zhao et al., 1999). Genes encoding conventional RBPs, those with canonical RNA binding domains, are predicted to account for >10% of the human genome (Mukherjee et al., 2019), and more unconventional RBPs, those without canonical RNA binding domains, keep being discovered (Castello et al., 2016; Perez-Perri et al., 2018).

Many RBPs have been implicated in human diseases, particularly in neurodegenerative diseases. Post-mitotic neurons cannot divide themselves out of trouble and therefore rely heavily on regulation to remain healthy and viable. In neurons, disruptions of RNA regulatory networks lead to serious pathological effects. For example, TDP-43 dysregulation in amyotrophic lateral sclerosis (ALS; Gao et al., 2018; Ling et al., 2013), fragile X mental retardation protein (FMRP) in fragile X syndrome (Penagarikano et al., 2007), and SMN1 in spinal muscular atrophy (SMA; Farrar and Kiernan, 2015) underscore how important proper RBP function is to neuronal health.

RBPs can be expressed ubiquitously across tissues (Gerstberger et al., 2014) or with tissue-specific and even organelle-specific patterns (Van Nostrand et al., 2020). An additional level of complexity arises from collaborations between RBPs (e.g.,

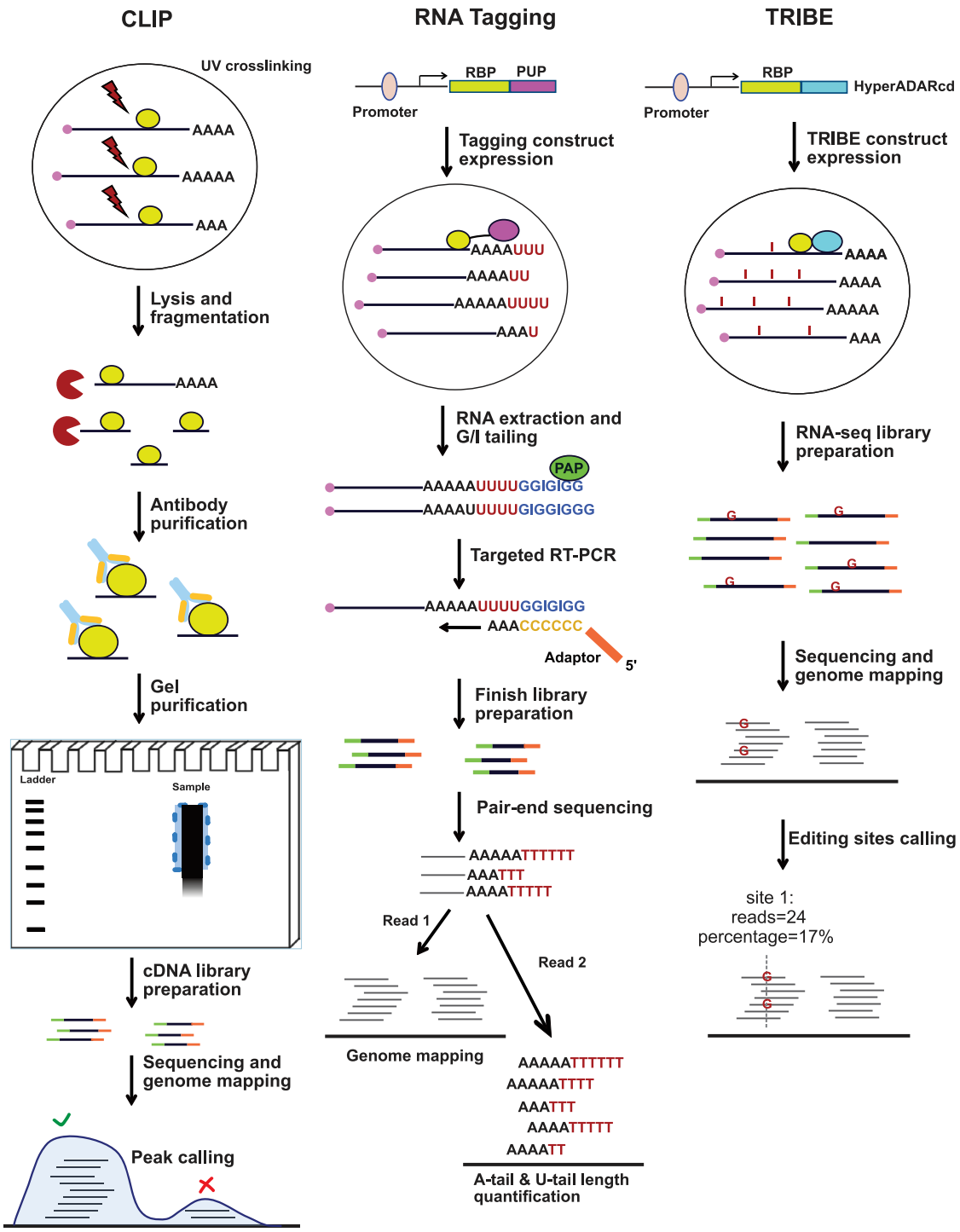

Figure 2. Workflow comparison of CLIP, RNA tagging, and TRIBE

From left to right: CLIP, RNA tagging, and TRIBE experimental workflows and simplified computational analysis processes.

CLIP: RBPs are crosslinked with their target RNAs with 254-nm UV light, followed by cell lysis. Cell lysates are treated by RNase to partially digest the RNA, leaving only the RNA fragments protected by RBPs. The RBP of interest and its protected RNA fragments are immunoprecipitated with the antibody specifically against the RBP. The RNP complexes are purified via gel electrophoresis and prepared into sequencing libraries. After sequencing, the sequencing reads are mapped to the genome and the CLIP peaks can be called based on the mapped reads through various computational pipelines. These CLIP peaks are considered the binding sites of the assayed RBP, and their heights reflect the binding affinity. Parallel regular RNA-seq libraries are usually prepared to serve as the input control for CLIP experiments (omitted in the figure).

(legend continued on next page)

RBP-X binding can be altered by the presence of RBP-Y; Smith and Valcárcel, 2000). Collaboration between RBPs underscores how important it is to define RBP targets *in vivo* and at the level of single-tissue and even single-cell resolution—in other words, avoiding whenever possible tissue and cellular heterogeneity.

RNA modification-based genetic methods were developed to examine RBP targets *in vivo* and also take advantage of the readily available and sensitive RNA-seq library preparation methods. The two first such methods, TRIBE (targets of RNA-binding proteins identified by editing; McMahon et al., 2016) and RNA tagging (Lapointe et al., 2015), use the RNA binding properties of a RBP to bring a RNA-modifying enzyme to specific transcripts, and then the enzyme modifies those transcripts. TRIBE uses the RNA editing enzyme ADAR as the RNA modifier (McMahon et al., 2016), whereas RNA tagging relies on the poly-U polymerase (PUP)-2 (Lapointe et al., 2015).

BRIEF OVERVIEW OF RNA EDITING APPLICATIONS

Because of its modular feature and RNA-specific editing activity, ADARs are a major molecular biology tool not only for RBP target identification (McMahon et al., 2016) but also for therapeutic ADAR-based recoding (Abudayyeh et al., 2019; Cox et al., 2017; Montiel-Gonzalez et al., 2013). APOBEC family proteins have been used less until recently, perhaps because they typically have ssDNA as well as RNA-editing activity; the presence of both activities can be confounding and problematic (Gaudelli et al., 2017).

When the ADARcd is fused to an RBP (in cases of RBP target identification) or to a target-guiding protein (in cases of therapeutic editing), substrate selection of the fusion protein is driven by the RBP or the target-guiding protein, and the ADARcd only serves as a deamination machine. Here, we summarize recent developments in RNA editing applications for target identification and therapeutic editing and provide some considerations for future directions.

TRIBE

In TRIBE, the *Drosophila* ADARcd is fused to the C terminus of an RBP. The latter drives the binding specificity of the fusion protein (McMahon et al., 2016), and the ADARcd deaminates adenosines near the RBP binding sites. The A-to-I editing marks are identified as A-to-G mutations by deep sequencing (Biswas et al., 2021b; McMahon et al., 2016; Rahman et al., 2018). Because many target RNA regions appear to lack canonical double-stranded or other well-defined structured regions, the ADARcd likely captures transient double-stranded regions due to the exploration of different structures by *in vivo* RNA. Moreover, most TRIBE editing sites are located within 100 bp of the RBP binding sites as determined by crosslinking immunoprecipitation (CLIP; see below). This indicates a relative lack of sequence specificity, which further suggests that TRIBE editing occurs in regions that fold transiently—TRIBE editing is not limited to regions that are structured by traditional criteria (McMahon et al., 2016).

TRIBE experiments can use cell- or tissue-specific expression of the RBP-ADARcd fusion protein. All library preparation protocols are compatible, which include mRNA-seq, total RNA-seq and/or nascent RNA-seq. After mapping to the genome, RNA positions with A-to-G "mutations" are detected when compared to genomic DNA or compared to control RNA with ADARcd-only expression. Editing sites are called using the following stringent operational criteria: (1) >80% A and 0% G at the site in the control samples; (2) an edited site must be covered by >20 reads with >10% G; (3) the site must be reproducible in all of at least two TRIBE replicates (Figure 2; McMahon et al., 2016; Rahman et al., 2018). Although looser TRIBE criteria have been used to meet specific needs (Alizzi et al., 2020; Singh et al., 2021; Xu and Rosbash, 2021), such stringent TRIBE criteria minimize false positive editing.

TRIBE was first applied to three RBPs, Hrp48, FMRP, and nonA (no-on-transient A) and could identify cell-specific targets in as few as 100 *Drosophila* circadian neurons (McMahon et al., 2016). TRIBE-identified targets were different for the three different RBPs and were also consistent with targets identified by CLIP (McMahon et al., 2016). Moreover, no editing was observed with expression of the ADARcd alone (i.e., editing required expression of a RBP-ADARcd fusion protein) (McMahon et al., 2016). Importantly, CLIP results with Hrp48-TRIBE expression in cell culture were identical to those with Hrp48 alone, suggesting that the addition of ADARcd residues to Hrp48 does not lead to major changes in Hrp48 specificity (McMahon et al., 2016). These results indicate that TRIBE editing can faithfully reflect the binding specificity of the RBP, even in small numbers of cells.

One remaining limitation of TRIBE was due to the substrate biases of the ADARcd, which include favoring bulged adenosines surrounded by double-stranded regions (Macbeth et al., 2005) as well as adenosines flanked by a 5′ A or U and a 3′ G (flanked by an A/U<u>A</u>G sequence) (Kuttan and Bass, 2012). These preferences promote base-flipping, the rate-limiting step of ADAR catalysis (Matthews et al., 2016). Put otherwise, the binding affinity of ADARcd to the favored U<u>A</u>G sequence and the unfavored G<u>A</u>C sequence is similar, but the base-flipping rate is

RNA tagging: expression constructs fuse the RBPs of interest and the PUP enzyme under specific promoters for expression in cells. The PUP enzyme adds poly-U tails after the original poly-A tails on the target mRNAs of the RBP. The first strand cDNAs are prepared by adding G/I mixed 3′ tails to RNAs using poly-A polymerase (PAP) in cell lysates and reverse transcribed with primers hosting a 3′ poly-C (complementary to the G/I tail) and poly-A (complementary to the poly-U tail) sequence. Sequencing libraries are then generated with the cDNA using standard procedures. RNA tagging libraries must be sequenced in paired-end fashion. The read 1 data allows mapping of the RNA fragment to the genome, and the read 2 information is used to determine the length of the U-tail—hence, the binding affinity of the RBP to the transcript.

TRIBE (HyperTRIBE): expression constructs fuse the RBPs of interest and the ADARcd for expression in cell culture using constitutive promoters or in specific cells using tissue-specific drivers. The fusion protein generates inosine editing marks on the target RNAs of the RBP, which are converted to guanosines in the final libraries (A-to-G changes). TRIBE samples can be prepared with various standard RNA-seq library preparation methods. TRIBE sequencing data are first mapped to the genome and then filtered for nucleotide sites that have new A-to-G mutations compared to genomic or wild-type (WT) RNA-seq data. Only the sites with sequencing coverage and A-to-G mutation rate above set threshold are considered editing sites, and the transcript as a target of the RBP. The STAMP experimental procedure is similar to TRIBE, with a different analysis pipeline for C-to-U mutations.

much higher for UAG; this results in a 2,000-fold difference in editing rate (Kuttan and Bass, 2012). These ADARcd biases almost certainly contributed to significant false negatives when TRIBE was originally compared to CLIP (McMahon et al., 2016), which also include the masking of some RBP binding sites by inefficient ADARcd-mediated editing.

HyperTRIBE

Point mutation screening identified a hyperactivating mutation in hADAR2 (Kuttan and Bass, 2012). hADAR2 carrying this E488Q mutation deaminates the favored UAG sequence 3-fold faster and even more importantly, the unfavored GAC 60-fold faster than wild-type (WT) hADAR2; this is due to the enhanced base-flipping ability of the mutant enzyme (Kuttan and Bass, 2012; Matthews et al., 2016).

By incorporating this hyperactivating mutation into dADARcd, HyperTRIBE (TRIBE version 2.0) dramatically increased the editing efficiency of the method, leading to the identification of many more editing sites and target RNAs. Many of these were not truly "new" sites, but were previously edited at sub-threshold levels with standard TRIBE (McMahon et al., 2016; Rahman et al., 2018; Xu et al., 2018). As a result, HyperTRIBE-identified targets are more consistent with CLIP results in cultured cells (i.e., less plagued with the false-negative problem). The higher editing efficiency of HyperTRIBE also decreased the sequencing depth requirements by approximately an order of magnitude, from 200 million reads (TRIBE) to 20–30 million reads (HyperTRIBE) with *Drosophila* libraries (McMahon et al., 2016; Rahman et al., 2018; Xu et al., 2018).

Analysis of the neighbor sequence context and local structure also indicated that HyperTRIBE effected a significant reduction in A/UAG sequence bias and double stranded structure (Markham and Zuker, 2008; McMahon et al., 2016; Rahman et al., 2018; Xu et al., 2018) (i.e., a reduced dependence on local context). We attribute these positive features of HyperTRIBE to the disproportional increase in editing efficiency on unfavored adenosines, which much more faithfully reflects the binding specificity of the RBP.

Since its initial publications, TRIBE and HyperTRIBE have not only been applied to other RBPs in *Drosophila* (Alizzi et al., 2020; Singh et al., 2021) but have also been successfully applied to other organisms, including malaria parasites (Liu et al., 2019), plants (Arribas-Hernández et al., 2021; Zhou et al., 2021), and mammalian cells (Table 1; Biswas et al., 2020; Herzog et al., 2020; Jin et al., 2020; Nguyen et al., 2020).

RNA tagging

As mentioned above, RNA tagging is another genetic tool to identify RBP targets *in vivo* (Lapointe et al., 2015). In this method, the RBP of interest is fused to a PUP, which adds poly-U sequences after the poly-A tail on the RBP targets. These poly-U tails and RBP targets can then be identified with next-generation sequencing.

PUP does not have the modular feature of ADAR and cannot be separated into a binding domain and a catalytic domain (Figure 1). The targets of RNA tagging therefore reflect both the specificity of the RBP and the specificity of the PUP. Moreover, RNA tagging has only been used in *Saccharomyces cerevisiae* to date, making it uncertain whether RNA tagging can be applied to higher eukaryotes.

APOBEC-based methods

TRIBE-like methods using the C-to-U editing enzyme APOBEC have been recently developed (Meyer, 2019). DART-seq (deamination adjacent to RNA modification targets) fused APOBEC1 with the binding domain of m^6A-binding protein YTHDF2 to profile m^6A modifications (Meyer, 2019). Comparing the editing data with results from m^6A immunoprecipitation experiments (miCLIP; Linder et al., 2015) validated DART-seq by showing that editing sites are enriched around m^6A modifications (Meyer, 2019). The DART-seq approach has been extended more recently to several other RBPs (STAMP, surveying targets by APOBEC-mediated profiling; Brannan et al., 2021), validating target identification by APOBEC-mediated editing as a general strategy.

Comparisons between genetic methods

With the growing numbers of available tools to identify RBP targets, it has become increasingly important to have comprehensive comparisons, ideally head to head in the same study. They will allow researchers to choose an optimal method (Figure 3; Table 2). Although direct comparisons of ADAR- and APOBEC-mediated editing using the same RBPs are in progress (data not shown), prior studies in *Drosophila* (McMahon et al., 2016) and *Arabidopsis* (Zhou et al., 2021) show that ADAR editing activity is higher than that of APOBEC. Only one paper thus far has directly compared HyperTRIBE (*Drosophila* ADAR E488Q) and DART-seq/STAMP (rat APOBEC1) in plants and found HyperTRIBE to be more efficient. Therefore, further studies are needed to determine why one enzyme may be better than another.

TRIBE/HyperTRIBE is to date the most widely adopted genetic method to identify RBP targets. Nonetheless, there remain limitations, which would benefit from further optimization. The double-strand structure and neighboring sequencing preference of the ADARcd (Markham and Zuker, 2008; McMahon et al., 2016; Rahman et al., 2018; Xu et al., 2018) still probably bias TRIBE/HyperTRIBE against RBP binding sites flanked by adenosines with suboptimal structures and neighboring sequences. Although this neighboring sequence and double-strand structure requirement are significantly reduced with HyperTRIBE (Xu et al., 2018), they are probably still the primary sources of HyperTRIBE false-negative errors. However, given the flexibility of RNA structure and the ability of HyperTRIBE to "reach" hundreds of base pairs from RBP binding sites, the chance seems remote that the entire mRNA transcript is single stranded and inaccessible to ADAR editing. These considerations also complicate quantitative interpretations; editing efficiency probably reflects these context effects as well as the affinity of RBP binding.

The reason that RNA tagging has not been applied to higher organisms is likely due to the concern that adding poly-U sequences to the 3′ end of mRNAs would cause mRNA degradation in human cells (Lim et al., 2014; Song and Kiledjian, 2007). Specialized RNA-seq library preparation (Figure 2) and analysis further limit the adoption of this technique (Lapointe et al., 2015) compared to other editing fusions (TRIBE/HyperTRIBE/DART-seq/STAMP).

Table 1. TRIBE experiments published

Organism	RBP	ADARcd	Hyperactive mutant (E488Q)	Negative controls	Cell type	RNA-seq library type	Mapped sequencing depth, read length	Reference
Drosophila melanogaster	Hrp48 (Hrb27C)	*Drosophila*	no	ADARcd, RBP mutant	S2 cells, neurons	mRNA	200 million, single-end 75 bp	McMahon et al., 2016
	dFmr1	*Drosophila*	no	ADARcd	S2 cells, neurons	mRNA	200 million, single-end 75 bp	McMahon et al., 2016
	NonA	*Drosophila*	no	ADARcd	S2 cells, neurons	mRNA	90 million, single-end 75 bp	McMahon et al., 2016
	NonA	*Drosophila*	no	ADARcd	S2 cells, neurons	nascent RNA	60 million, single-end 75 bp	McMahon et al., 2016
	Hrp48 (Hrb27C)	*Drosophila*	yes	ADARcd	S2 cells, neurons	mRNA	20 million, single-end 75 bp	Xu et al., 2018
	Thor (d4E-BP)	*Drosophila*	yes	ADARcd	S2 cells	mRNA	20 million, single-end 75 bp	Jin et al., 2020
	Fne	*Drosophila*	no	ADARcd	S2 cells	mRNA	90 million, single-end 180 bp	Alizzi et al., 2020
	Ataxin-2	*Drosophila*	no	RBP mutants	Neurons	mRNA	25 million, paired-end 125 bp	Singh et al., 2021
	Ythdf	*Drosophila*	yes	ADARcd	S2 cells	mRNA	10 million, paired-end 75 bp	Worpenberg et al., 2021
	Fmr1	*Drosophila*	yes	ADARcd	S2 cells, larval brain	mRNA	10 million, paired-end 75 bp	Worpenberg et al., 2021
Homo sapiens	MCP	*Drosophila*	yes	mCherry ADAR, loss of target	U2OS cells	mRNA	40 million, paired-end 150 bp	Biswas et al., 2020
	TDP-43	Human	yes	ADARcd, mutant RBP	HEK293T cells	mRNA	40 million, single-end 75 bp	Herzog et al., 2020
	4E-BP	Human	yes	ADARcd	PC3 cells	mRNA	40 million, single-end 75 bp	Jin et al., 2020
	YTHDC-1	*Drosophila*	yes	vector control	MOLM-13 cells	mRNA	40 million, paired-end 100 bp	Cheng et al., 2021
	MSI2	*Drosophila*	yes	ADARcd, mutant RBP	Hematopoietic stem cells, human AML cells	mRNA	40 million, paired-end 100 bp	Nguyen et al., 2020
Mus musculus	MCP	*Drosophila*	yes	mCherry ADAR, loss of target	MEFs	mRNA	50 million, paired-end 150 bp	Biswas et al., 2020
Rattus novergicus	TDP-43	Human	yes	ADARcd	Rat cortical neurons	mRNA	40 million, single-end 75 bp	Herzog et al., 2020
Nicotiana benthamiana	Ubp1c	*Drosophila*	yes	ADARcd	Plant cells	mRNA	40 million, paired-end 150 bp	Zhou et al., 2021
Oryza sativa	Ubp1c	*Drosophila*	yes	Transgene negative parent	Plant cells	mRNA	40 million, paired-end 150 bp	Zhou et al., 2021
Plasmodium falciparum	Dis3	*Drosophila*	no	WT sequencing, RBP knockdown	Malaria parasites	mRNA	paired-end 75 bp	Liu et al., 2019

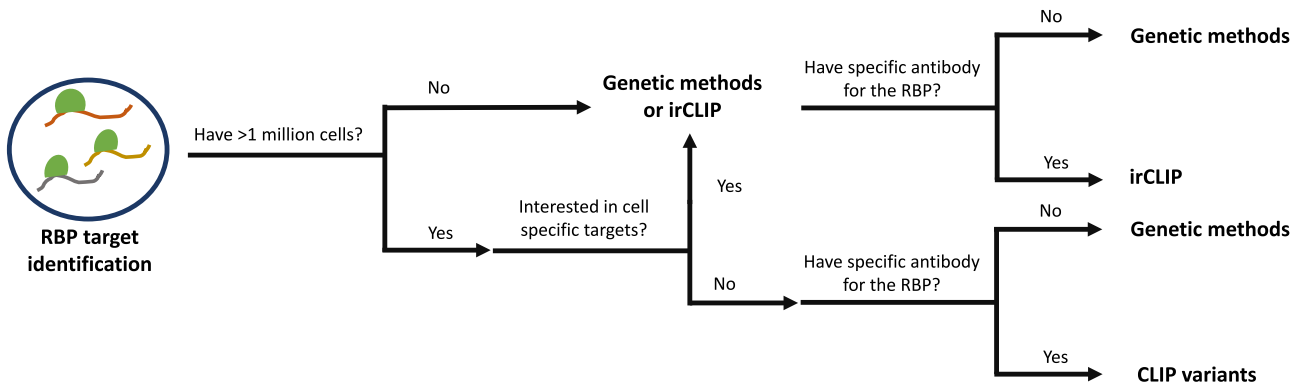

Figure 3. Flowchart for RBP target identification method selection
Considerations when selecting a method (bold text) to study your RBP of interest (green circles) include the amount of starting material, the requirement of cell subtype specificity, and the presence of specific antibodies (text above arrows).

The ability of APOBEC to edit ssRNA also creates potential limitations as a target identification tool. Both DART-seq and STAMP fuse the RBP of interest with the entire APOBEC1 enzyme. Like PUP, the editing moiety of APOBEC1 has not been separated from its RNA binding domain (Figure 1). Moreover, APOBEC1-mediated editing requires its RNA binding partner ACF and formation of APOBEC1 dimers (Lau et al., 1994; Navaratnam et al., 1998). Therefore, the specificity of STAMP as dictated by a fused RBP may be compromised by APOBEC1-ACF-mediated editing. There appears to be considerable APOBEC-only editing. Although this is successfully subtracted by computational pipelines (Brannan et al., 2021; Meyer, 2019), such background RNA editing, including background ssDNA editing, may prove more problematic in other paradigms, including in transgenic animals. To accommodate both ssDNA and RNA background/off-target editing, it is better to use both genomic DNA and control RNA as references to reduce false positives.

APOBEC editing in coding regions can also create premature stop codons, by converting CAA, CAG, and CGA into UAA, UAG, and UGA stop codons, respectively. This could lead to false-negative errors and also affect cell or animal viability, including through their effects on mRNA stability via nonsense-mediated decay (NMD; Brogna and Wen, 2009). In this context of viability, the enhanced efficiency of APOBEC-mediated editing relative to ADAR-mediated editing may be a liability (Brannan et al., 2021). Finally, endogenous APOBEC1 editing sites are predominantly A/U flanked (Rosenberg et al., 2011), which may confound DART-seq/STAMP.

One feature shared by these different tools is the requirement of expressing recombinant RBP-editing enzyme fusion proteins (Table 2). The simplest initial approach to do this is via overexpression. This may cause ectopic RBP binding to non-physiological targets, resulting in false positive errors as well as possible deleterious effects. One way to avoid overexpression is to introduce via CRISPR a rapamycin-inducible FKBP/FRB dimerization system (Komatsu et al., 2010; Patury et al., 2009). The small FKBP/FRB epitope can be CRISPR knocked in to the RBP, in which case, the ADARcd tagged with FRB/FKBP can still be overexpressed in the cells of interest. In this system, no editing will occur until the RBP and ADARcd are brought into proximity with rapamycin.

Another shared limitation is the failure to identify the exact RBP binding sites/motifs on target RNAs. Since RBP binding probably blocks editing enzymes from modifying the precise binding site, most TRIBE and STAMP editing sites are located near but not within the RBP binding sites (Brannan et al., 2021; McMahon et al., 2016). Nevertheless, recent studies indicate that it is feasible to identify the RBP binding sites/motifs by computationally searching for enriched motifs within specific regions of the TRIBE-identified targets (Nguyen et al., 2020; Singh et al., 2021). A similar approach has recently been taken with APOBEC-mediated editing (Brannan et al., 2021; Meyer, 2019).

Biochemical methods to identify RBP targets (RNA immunoprecipitation [RIP] and CLIP variants)

Before genetic methods were developed, immunoprecipitation from cell or tissue lysates was the most widely adopted approach to determine RBP targets (Keene et al., 2006; Ule et al., 2003). RIP is similar to chromatin immunoprecipitation (ChIP) and begins with an antibody against a specific RBP of interest (Sundararaman et al., 2016). The antibody enriches for complexes containing the RBP, and a mild lysis buffer is generally used to preserve complex integrity (Keene et al., 2006). The RNA content of the complexes can then be profiled by microarray or sequencing methods. RIP was the first tool to identify transcriptome-wide RBP targets (Gagliardi and Matarazzo, 2016). Although RIP is reliable for assaying stable RNA-protein interaction, RNPs with weaker interactions may dissociate and even reassociate during lysis, leading to both false-negative and false-positive errors (Mili and Steitz, 2004; Riley et al., 2012). RIP experiments therefore require many replicates due to their relatively low reproducibility rate (60%–75% at transcript level; Khalil et al., 2009). This complicates both experimental logistics and data interpretation, reducing confidence in bona fide targets.

In vivo crosslinking can be used to create a covalent interaction between the RBP and its target RNAs. CLIP with UV light was for this reason introduced to overcome the stability limitations of RIP (Ule et al., 2003). CLIP development revolutionized the study of RBPs, and successors to the original CLIP protocol

Table 2. Comparisons between RBP target identification methods

	TRIBE	RNA tagging	STAMP/DART-deq	RIP	CLIP
Recombination protein expression	yes	yes	yes	no	no, except when epitope tag used
High-quality antibody required	no	no	no	yes	yes
Material needed	~100 cells	75 μg total RNA	as low as single cell	millions of cells	1 million cells
Library preparation	standard RNA-seq	special protocol	standard RNA-seq	standard RNA-seq	special small RNA protocol
Abundance bias	low	unknown	unknown, likely low	high	high
Local sequence bias	yes, UAG	unknown	yes, A/U flanked	no	yes, U-rich regions
Time for covalent mark	~24 s	seconds	seconds	N/A	instant
Target identification resolution	transcript/region	transcript	transcript/region	transcript	binding site

have used different crosslinkers as well as modified protocols to take advantage of the continuing advances in next-generation sequencing (NGS) technology. Once crosslinked *in vivo*, RBPs and their bound RNA fragments are then purified by immunoprecipitation. Because of the UV-mediated covalent association between RNAs and proteins, conditions of cell lysis and immunoprecipitation can be much more stringent in CLIP than in RIP (Ule et al., 2003). Once immunoprecipitated, the crosslinked protein RNA complex is run on a denaturing gel. Bands corresponding to the size of the complex are isolated, after which multiple steps are used to remove the protein and introduce RNA-seq adapters to the isolated, crosslinked RNA. CLIP coupled to high-throughput sequencing (HITS) is referred to as HITS-CLIP (Figure 2; Chi et al., 2009; Darnell, 2010). HITS-CLIP and its variants have become the predominant mode of *in vivo* RBP target determination (see below). For in-depth reviews of CLIP methods and discoveries, we point the reader to several reviews (Hafner et al., 2021; Lee and Ule, 2018).

Modified CLIP protocols have been developed over the years, most notably PAR-CLIP (Hafner et al., 2010), iCLIP (König et al., 2010), and eCLIP (Van Nostrand et al., 2016, 2017). To increase the UV crosslinking yield, PAR-CLIP incubates cells with photoactivable 4-thiouridine before crosslinking (Hafner et al., 2010). In addition, 4-thiouridine base pairs with guanosine during reverse transcription and thereby introduces T-to-C mutations into the sequencing results; this provides a second layer of confidence to the CLIP peaks (Hafner et al., 2010). iCLIP (König et al., 2010) identifies RBP binding sites with single-nucleotide resolution by using the truncation products of reverse transcriptase at the crosslinking site. eCLIP evolved from iCLIP and is the most recent CLIP method. eCLIP has the advantages of being radiation independent, with a shortened experimental procedure and increased library preparation efficiency (Van Nostrand et al., 2016).

The ENCODE 3 (Encyclopedia of DNA Elements) project has published target site eCLIP data for 150 RBPs (Van Nostrand et al., 2020) from 2 human cell lines (HepG2 and K562). By combining the eCLIP data with information regarding alternative splicing, changes in gene expression, and localization as well as knockdown, Bind-N-Seq, and imaging assays (Van Nostrand et al., 2020), ENCODE 3 also associates RBPs with specific post-transcriptional regulatory processes and provides information on one-third of all known human RBPs (356 RBPs studied in total). Importantly, the remaining 90% of known RBPs and other unconventional RBPs (Gerstberger et al., 2014) remain difficult to interrogate due in part to a lack of high-quality, CLIP-compatible antibodies.

Comparisons between editing and CLIP/RIP

Although CLIP and RIP optimization is constantly progressing, there are still intrinsic limitations even to the most up-to-date CLIP and RIP protocols. Immunoprecipitation requires dependable and specific anti-RBP antibodies, yet different excellent antibodies against the same RBP can still show divergent results (Khalil et al., 2009; Lambert et al., 2014). Moreover, antibody-related artifacts and high background strongly influence experimental outcomes (Friedersdorf and Keene, 2014). Most important, CLIP and RIP require millions of cells per replicate, presumably because of inefficiency at multiple steps (Moore et al., 2014); only irCLIP is exceptional, but still with a minimal requirement of 20,000 cells (Zarnegar et al., 2016). It is notable that there are cell-specific CLIP protocols similar in principle to TRIBE that express tagged RBPs in mammalian brain tissues (Hwang et al., 2017; Yuan et al., 2018). However, it is still uncertain whether these approaches have sufficient signal:noise to assay very sparse cell types.

TRIBE, in contrast, has made investigating RBP targets in scarce neurons feasible (Herzog et al., 2020; McMahon et al., 2016; Rahman et al., 2018) and prioritizes this ability to perform cell-specific assays. This is also because RBP targets can vary substantially across different cell types (Brannan et al., 2021; McMahon et al., 2016), especially in neurons. This risks making biochemical approaches to assaying heterogenous tissues confounding, if not misleading. Although TRIBE cannot identify precise RBP binding sites at nucleotide-level resolution like CLIP, TRIBE can still identify an approximate binding region; this is because TRIBE editing sites are often located within 100 bp of CLIP peaks of target transcripts, as mentioned above. These regions, together with simple analysis tools such as MEME Suite, can be analyzed to determine precise binding motifs along with the many available *in vitro* methods (e.g., SELEX, RNA Bind-N-Seq). Lastly, it has been shown that even after normalization, CLIP data correlate strongly with RNA abundance. The correlation is much less strong

with TRIBE data (Biswas et al., 2020), a distinction that is consistent with the better performance of TRIBE at identifying low-abundance targets (Arribas-Hernández et al., 2021).

Another advantage of the genetic methods is their applicability to RBP identification in most research labs; there is no requirement for antibodies at all, let alone reliable and high-affinity antibodies. Moreover, TRIBE (and DART-seq/STAMP) use regular RNA-seq library protocols, which save time, labor, and even money compared to RIP and CLIP protocols. Because of the low percentage of usable reads, CLIP libraries typically require >100 million reads of sequencing coverage. Even the advanced eCLIP method generates only ~30% of useful reads and requires >50 million reads (Van Nostrand et al., 2016). In contrast, HyperTRIBE editing sites have been found with 20 million reads in *Drosophila* (Xu et al., 2018) and 40 million reads in mammalian samples (Herzog et al., 2020).

To measure RBP binding preference and decouple it from transcript abundance, one needs to normalize the abundance of the CLIP tags/TRIBE editing sites to the transcriptome distribution. In mammals, the annotated transcriptome distribution is 50%, 40%, and 10% for the coding sequence (CDS), 3′ untranslated region (UTR), and 5′ UTR, respectively (Piovesan et al., 2019). Taking FMRP as an example, ~60%–70% of the CLIP tags or TRIBE signals locate to the CDS (Darnell et al., 2011; McMahon et al., 2016), but there is still no obvious enrichment of any mRNA region after normalization (i.e., there is no real binding preference). This idea is supported by the identification of FMRP binding motifs in both CDS and 3′ UTR regions (Anderson et al., 2016). Importantly, normalization is intrinsic to TRIBE/STAMP data; editing is always expressed as a percentage of sequencing reads. In addition, the HyperTRIBE editing level was found to be independent of the transcript expression level (Nguyen et al., 2020). CLIP data, in contrast, rely on a separate input of RNA-seq data for normalization.

TRIBE and STAMP more reliably identify candidate targets. Although these targets are highly correlated with CLIP results, they often contain additional novel targets (Brannan et al., 2021; Xu et al., 2018). Important to evaluating all candidate lists is further validation by other approaches. In one TRIBE example, the high target reproducibility increased confidence in the identification of novel transcripts and allowed the identification of candidate pathways previously invisible in the much longer list of CLIP targets (Herzog et al., 2020).

MS2 TRIBE directly compares signal and noise from CLIP and TRIBE

Integrating CLIP and HyperTRIBE data can define RNA binding sites as well as a repertoire of high-confidence RBP targets. As noted above, each approach has its own limitations, such as antibody specificity for CLIP and fusion protein expression for HyperTRIBE. In this context, it is unclear how to interpret RNA targets found by only one of the two techniques. A true positive control can begin to address this TRIBE versus CLIP issue. The MS2 capsid protein (MCP) and its RNA target, the MS2 binding site (MBS), constitute a widely used system for single-molecule RNA imaging and have previously been shown to interact with high affinity and specificity within both prokaryotic and eukaryotic cells (Bertrand et al., 1998; Tutucci et al., 2018). The comparison of MCP Hyper-TRIBE targets with MCP-CLIP results indicated that both techniques can identify a true positive control (i.e., an endogenous RNA engineered to contain the MBS stem loops within its 3′ UTR). However, control experiments also identified antibody-dependent background as well as pan-CLIP RNAs—namely, transcripts that appeared across multiple CLIP experiments against different RBPs (Biswas et al., 2020).

Although antibody specificity had been previously shown to be important for accurate CLIP results (Sundararaman et al., 2016), background signal was still evident even with highly specific antibodies (Biswas et al., 2020). Moreover, the aforementioned MCP/MS2 control experiments showed that HyperTRIBE background editing was 5–10 times lower than CLIP background when no true positive target was present (Biswas et al., 2020). Also, the MS2-TRIBE experiments were performed using transiently transfected MCP-ADAR expressed from a minimal ubiquitin C promoter. In this case, the higher levels of transiently transfected RBP-ADAR did not lead to increased off-target editing after the subtraction of editing control values, which argues that RBP overexpression at a modest level may not cause novel or spurious interactions (Biswas et al., 2020). Questions remain as to the cause of background CLIP peaks: some background may be caused by non-specific antibody interactions, and other background sources (Van Nostrand et al., 2016) may include highly expressed RNAs (Van Nostrand et al., 2017).

Recommendations for RBP target identification tool selection

Selecting the proper tool for RBP target identification depends on the question of most interest: is it defining the RBP targets or the precise RBP binding sites? The aforementioned genetic tools are preferable for the former, whereas CLIP variants and especially eCLIP are more appropriate for the latter. When it is difficult to express the recombinant editing proteins *in vivo* and there are high-quality antibodies against the RBP available, CLIP would be the best option. It is also worth noting that the cost of CLIP (and its variants) can be significantly more because of its requirement for higher sequencing coverage and high-quality antibodies (Figure 3).

Among the genetic tools, TRIBE/HyperTRIBE has been shown to be applicable to different animal and plant models. Organism applicability of STAMP/DART-seq is currently unknown, almost certainly because it is so new. However, we expect it to be similar to TRIBE. Another consideration for these genetic tools is the type of RNA-seq experiment to perform (i.e., performing nascent RNA-seq or nuclear RNA-seq for nuclear RBPs and mRNA-seq for cytoplasmic RBPs would be ideal). Nuclear RBP-TRIBE likely deposits many editing sites on introns, which will not be detected by mRNA-seq. This distinction may explain some previously observed target number discrepancies between HyperTRIBE and CLIP data (Cheng et al., 2021).

BROADER AND NEW APPLICATIONS OF EDITING-BASED METHODS

m^6A profiling

In addition to the recoding of RNA nucleotides, chemical RNA modification has been implicated in nearly all aspects of RNA

regulation (Meyer and Jaffrey, 2017; Zaccara et al., 2019). The most abundant of these modifications is m^6A and methods to define m^6A sites across the transcriptome have been developed. These methods initially began with antibody-based methods, meRIP-seq (Meyer et al., 2012), m^6A-seq (Dominissini et al., 2012), and meCLIP (Linder et al., 2015), and have more recently exploited the antibody-independent approaches APOBEC1-based DART-seq (Meyer, 2019) and ADAR-based m^6A-TRIBE (Worpenberg et al., 2019). These methods now allow profiling with low input samples (Meyer, 2019; Worpenberg et al., 2021) and may in the future allow for single cell profiling of m^6A modifications (future directions).

MS2 TRIBE to identify nascent RNA-RNA interactions

Signal amplification technologies have been a hallmark of single-molecule imaging; for example, processes such as transcription (Janicki et al., 2004), RNA localization (Bertrand et al., 1998; Park et al., 2012), translation (Halstead et al., 2015; Wu et al., 2016), and RNA decay (Horvathova et al., 2017) have been observed by localizing multiple fluorescent proteins into a diffraction limited spot. The co-localization of enzymatic reactions has also been used to track RNA, most recently by expressing biotin ligase or ascorbate peroxidase enzymes at different organelle surfaces (Fazal et al., 2019; Jan et al., 2014; Williams et al., 2014). Similarly, localizing multiple ADAR molecules (~130) to a single, highly transcribed mRNA transcription site using the 24xMS2 stem loops causes high editing levels of the target mRNA, and other nearby mRNAs are also edited, albeit at lower frequencies (Biswas et al., 2020). As expected, there is no RBP-ADAR at high concentration and also almost no editing above background in the absence of the mRNA scaffold. This indicates that there are not only the many direct RBP-ADAR interactions with RNA (in *cis*) but also less expected in *trans* interactions: apparently the >100 copies of an RBP at a given locus can "reach out" to edit nearby RNAs. This supports the notion that nascent RNAs from nearby chromosomes are intimately associated with the locus and further suggests that protein factors may even be able to exchange from one transcription site to another (Shah et al., 2018). It is unclear whether phase separation is relevant to such trans editing, but this possibility, even at limited distances between loci, may be in line with recent observations (Guo et al., 2019; Hnisz et al., 2017). In any case, these observations indicate that MS2-TRIBE can be used as a "one vs all" method analogous to 4C, to map RNA contacts originating from a given transcription site. Moreover, RNA-RNA contacts from a single locus are mapped across the entire transcriptome and integrated over time (see below). MS2-TRIBE additionally brings the advantages of low input requirements (1,000 times less than 4C), simplified protocols, and high sequencing efficiency, which means that RNA-RNA contacts can be readily defined within small cell populations such as the nervous system or the hematopoietic system.

Time-dependent mark accumulation

The use of enzymes to mark discrete RNA targets can provide additional information. By tethering enzymes such as ADAR or APOBEC to a favorable target RNA *in vivo*, the enzymes can catalyze multiple reactions and cause a time-dependent accumulation of covalent marks on their substrates (i.e., A-to-I or C-to-U changes). MCP-HyperTRIBE, for example, shows such an increase in editing for both endogenous and reporter RNAs with the MS2 system (Biswas et al., 2020; Rodriques et al., 2021). Editing increases can then be used to infer RNA lifetime (Rodriques et al., 2021), suggesting that the degree of editing is a readout of the RBP dwell time. Likewise, PUP exhibits a time-dependent increase in tail length (Medina-Munoz et al., 2020), which correlates with target significance. Although further work is needed, it is likely that the number of enzymatic marks correlates with the affinity and off-rate of a protein-RNA complex, with some variability due to local structure and sequence context.

Translational profiling

Ribo-STAMP was developed very recently to investigate translation *in vivo*; the expression of a fusion protein between a ribosomal protein and APOBEC was able to induce translation-related editing sites in cultured cells (Brannan et al., 2021). RiboTRIBE is similar (Xu and Rosbash, 2021), and translation-dependent editing methods will be especially useful in systems with small numbers of cells for which biochemical techniques such as ribosome profiling are impractical. It will be especially interesting to see the extent to which Ribo-STAMP works in transgenic animals. This is likely to be highly dependent on the compatibility of Ribo-STAMP fusion protein with ribosome assembly as well as the signal:noise of the method. In this context, the extensive 3′ UTR RiboSTAMP signal indicates that some of the editing events may not reflect conventional protein synthesis. Moreover, the effect of Torin on APOBEC-mediated editing may reflect the impact of the drug on ribosome assembly rather than directly on protein synthesis, and it would also be reassuring to see enhanced editing of some exceptional Torin-upregulated transcripts.

Future possibilities

As described above, ADAR-based methods have begun to be widely adopted by several labs working in different organisms, and we anticipate that APOBEC and PUP-based methods will continue to gain traction and popularity. Given the substrate limitation and biases of both TRIBE and DART-seq/STAMP, it is tempting to consider a method that uses both enzymes to label the targets of one RBP. Fusing the ADARcd and APOBEC1 to the same RBP would allow cross-referencing between A-to-G and C-to-T editing sites on the same transcripts. Double editing would also provide a strong validation of bona fide RBP targets. In a similar vein, fusing the ADARcd and APOBEC1 to two different RBPs should allow for their study with one assay and enable the investigation of RBP collaboration/competition by examining the synergy/exclusion of the two types of editing events. Further integration with PUP-2 would even allow for triple editing within the same cells.

Immunoprecipitation-free enzymatic methods provide exquisitely high sensitivity and can use low-input materials. The recent combination of editing tools and single-cell RNA-seq is a particularly exciting development (Brannan et al., 2021; Einstein et al., 2021). We anticipate further development of single-cell RBP target profiling methods, including the use of ADAR and PUP

enzymes, allowing for the interrogation of RBP targets over the cell cycle or down a differentiation pathway.

Another interesting direction is to examine RNA localization at subcellular locations. This idea was first described with RNA tagging (Medina-Munoz et al., 2020) and is functionally similar to APEX-seq (Fazal et al., 2019). In this method, a protein is specifically anchored to an organelle or subcellular site. For example, the endoplasmic reticulum (ER) is tethered with the poly-A binding protein (PABP) and the PUP enzyme (Medina-Munoz et al., 2020). The inclusion of PABP provided the necessary RNA binding affinity for the editing activity of the RNA modifier. However, unlike APEX, which is based on streptavidin precipitation and requires large amounts of RNA, RNA localization with RNA-modifying enzymes can be achieved with a very limited number of cells using RNA-seq library protocols. Moreover, if PUP, ADAR, and APOBEC were anchored to three different subcellular locations within the same cell, these locations and their associated transcripts could be profiled simultaneously.

Now that multiple, antibody-independent methods are available to profile RBP targets at modest sequencing depth and low cost, it may be of interest to expand upon prior large-scale community-wide profiling projects (e.g., ENCODE; Van Nostrand et al., 2020). High-throughput synthesis of RBP-ADAR or RBP-APOBEC plasmid libraries could allow for a reusable publicly available resource to profile targets of both well-characterized and non-canonical RBPs in multiple cell lines. Initial experiments should use HepG2 and K652 cells, as 150 RBPs have already been profiled in these lines with eCLIP (Van Nostrand et al., 2016, 2020). The remaining ~90% of RBP candidates could then be profiled. This would provide a comprehensive map of RNA regulation, which could also incorporate total RNA counts within a given cell line. One can imagine that, when combined with quantitative methods such as molecular counting of RNA-seq data and single molecule RNA (smRNA) fluorescence *in situ* hybridization (FISH), RBP occupancy of all cellular transcripts could be envisioned.

Therapeutic RNA editing using ADAR

The dsRNA requirements for ADAR editing provide enticing opportunities for RNA therapeutics. Concerns about DNA base editing include heritable off-target mutations (Zhang et al., 2015) and chromothripsis due to on-target CRISPR-Cas9-based genome editing (Leibowitz et al., 2021). RNA base editing, in contrast, may be a safer alternative due to its non-heritable nature. To date, RNA base editors can be broadly grouped based on whether they recruit endogenous ADAR to the site of interest or whether they require an ADAR fusion protein (MCP-ADAR, λN-ADAR, SNAP-ADAR, Cas9-ADAR) to facilitate recruitment.

Tethering antisense oligonucleotides to ADAR has been shown to direct editing and with higher efficiency if the targeted adenosine forms an A-C mismatch with the antisense oligonucleotide. The mismatch promotes base flipping of the bulged adenosine (Matthews et al., 2016). Antisense oligonucleotides can thus generate an ideal RNA editing substrate at their hybridizing site (Merkle et al., 2019; Qu et al., 2019). Whereas long, linear antisense oligonucleotides have been used (Katrekar et al., 2019), RNAs that also contain the ADAR recruiting region of GluR2 mRNA (Fukuda et al., 2017; Wettengel et al., 2017) or a U6+27 cassette have a higher efficiency (Katrekar et al., 2021). Degradation of these short, uncapped, non-polyadenylated RNAs may limit their half-life in cells and therefore limit overall editing efficiency. To overcome this issue, circularized ADAR-recruiting RNAs were generated by flanking the targeting antisense RNAs with twister ribozymes (Katrekar et al., 2021); they undergo autocatalytic cleavage and are circularized by endogenous RtcB RNA ligase (Litke and Jaffrey, 2019). In addition to longer lifetimes, autocatalytic circularized ADAR-recruiting RNAs have fewer transcriptome-wide off-target editing sites and more effectively edit their target transcripts *in vivo* than their linear counterparts (Katrekar et al., 2021).

Although effective recruitment of endogenous adenosine deaminase can correct a wide range of genetic diseases and disorders (~20% of pathogenic SNPs in ClinVar; Cox et al., 2017), the editing efficiency of individual nucleotides has been modest. Higher efficiency has been observed with ADAR fusion constructs, which tether adenosine deaminase more effectively to a target RNA. For example, RNA chemically modified with benzyl guanine can be covalently linked *in vivo* to a SNAP-tagged ADAR protein and lead to higher editing efficiency (Vogel et al., 2018).

Noncovalent recruitment also increases editing efficiency. By engineering the targeting RNA to contain a stem loop structure (MS2 or Box B) and fusing ADAR to the respective capsid protein (MCP or λ-phage N), editing efficiencies can likewise be increased (Azad et al., 2017; Katrekar et al., 2019; Montiel-González et al., 2013, 2016). Along these same lines, CRISPR-Cas9, CRISPR-Cas13, or Cas7-11 guided ADARcds can correct pathological mutations at the RNA level in human cells (Abudayyeh et al., 2019; Cox et al., 2017; Özcan et al., 2021). In addition, new ADAR variants have been generated to allow for both adenosine and cytosine deamination from the same enzyme (Abudayyeh et al., 2019).

However, significant obstacles remain before these approaches can be used in human clinical trials (Marina et al., 2020). The large size of ADAR fusions makes viral packaging challenging (Katrekar et al., 2019), and a balance between high ADAR activity and off-target editing must be optimized (Katrekar et al., 2019). To this end, several approaches, including further mutagenesis of ADAR and Cas9/Cas13 are under way (Abudayyeh et al., 2019; Cox et al., 2017; Katrekar et al., 2021; Thuronyi et al., 2019). Even with optimization, however, editing by ADAR fusions in mice has been quite inefficient (<10%, Katrekar et al., 2021). Although the situation requires additional study and editing efficiency will certainly continue to improve, experience to date underscores the distinction between human therapeutics and biological tool development. Highly reproducible efficient ADAR-catalyzed editing in tissue culture and in model organisms is having a scientific impact, but such editing is only a necessary first step and far from sufficient for the full realization of its therapeutic potential in humans.

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DECLARATION OF INTERESTS

W.X. and M.R. declare that a PCT patent application (PCT patent application no. PCT/US2016/054525) on the TRIBE technique has been filed. The authors declare no other competing interests.

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